

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

September 2024 – December 2024

BU Spark!

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Real-Time Analytics Pipeline | *Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose* [Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

Crime Analytics & Forecasting Platform | *Python, Streamlit, Prophet, Scikit-learn, GeoPandas* December 2024

- Built crime analytics platform processing 500K+ Boston crime records with Prophet time-series forecasting and DBSCAN clustering for hotspot identification across 12+ police districts
- Implemented advanced data pipeline with seasonal decomposition and confidence intervals, generating 2-month crime predictions with train/test validation
- Developed interactive dashboard using Streamlit and Plotly, creating 121 crime-type heat maps with Folium and real-time geospatial visualizations

Audio Transcription Platform | *Java Spring Boot, Python FastAPI, React, PostgreSQL*

September 2025

- Built microservices platform with Spring Boot API, Python FastAPI service, and React frontend, implementing async audio processing with OpenAI Whisper and real-time status polling
- Designed PostgreSQL database with 4 related tables, Flyway migrations, and optimized indexing, supporting multiple audio formats with comprehensive error handling
- Deployed containerized application using Docker Compose with inter-service communication, CORS configuration, and health monitoring endpoints

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS

Data Processing & Streaming: Apache Kafka, Apache Flink, Pandas, NumPy, PySpark, Stream Processing

Databases & Storage: PostgreSQL, InfluxDB, Time-Series Databases, Data Warehousing, ETL/ELT Pipelines

Machine Learning & Analytics: Scikit-learn, Prophet, Time-Series Forecasting, Statistical Analysis, DBSCAN Clustering

Data Visualization: Plotly, D3.js, Streamlit, Folium, Geospatial Analysis, Real-Time Dashboards

Infrastructure & Tools: Docker, Docker Compose, Git, CI/CD, Data Pipeline Orchestration, WebSocket Streaming